

Engineering Mathematics III (ENGE702)

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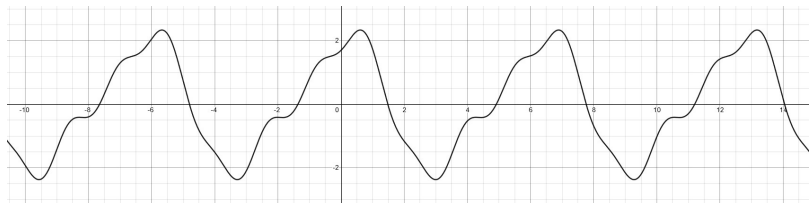
Lecture 18: Fourier Series

Fourier Series

Fourier series (and Fourier transforms) are needed for some methods of solving PDEs that we will see in later weeks. They are named after Jean-Baptiste Fourier (1768-1830), a French physicist and mathematician. Assume we have a periodic function

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

A periodic function is a function which 'repeats' itself after a finite length of time, eg:



Mathematically, this means there is a number p such that $f(x + p) = f(x)$ for every x . If this is the case, p is called a *period* of $f(x)$. The smallest such period is called the fundamental period.

Fourier Series

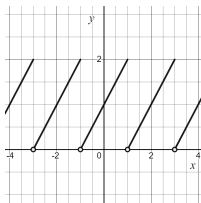
Let us assume that our periodic function

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

is piecewise continuous and that, at discontinuities, finite right and left hand limits exist. Let us also assume that, over a period T

$$\int_0^T |f(x)|^2 dx < \infty$$

that is, f is square integrable over a period. In the example below, $T = 2$. Then the Fourier series for f exists. It is an (infinite) sum of scaled $\sin$ and $\cos$ functions that converges to f in the sense that, if $S_n(x)$ is the sum to n terms of the Fourier Series, then $\lim_{n \rightarrow \infty} \int_0^T |S_n(x) - f(x)|^2 dx = 0$. This is convergence in the mean.



Fourier Series

Fourier analysis can be done over the complex numbers too, but we will restrict to using real trigonometric functions (sin and cos functions).

The sin and cos functions are basic periodic functions.

- ▶ $\sin(x)$ and $\cos(x)$ have a period of 2π .
- ▶ $\sin(2x)$ and $\cos(2x)$ have a period of π .
- ▶ In general, $\sin(nx)$ and $\cos(nx)$ have a period of $\frac{2\pi}{n}$, for $n \geq 1$.

Fourier Series

A function $f(x)$ of period 2π , for which a Fourier series exists, can be represented as follows:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

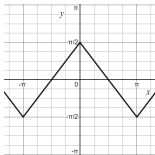
where:

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

$n \geq 1$. These are the Euler formulae, for a function with period 2π .



Identities

- ▶ $\frac{1}{L} \int_{-L}^L \sin\left(\frac{m\pi}{L}x\right) \sin\left(\frac{n\pi}{L}x\right) dx = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$
- ▶ $\frac{1}{L} \int_{-L}^L \cos\left(\frac{m\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) dx = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$
- ▶ $\int_{-L}^L \sin\left(\frac{m\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) dx = 0$

for integers m and n . These identities may be proved by transforming the integrands from products to sums. See the textbook (Chapter 11.1).

The theory of Fourier series depends on the orthogonality of the trigonometric system.

Set $L = \pi$ and use the following identities to prove the Euler formulae on the previous slide.

Odd and Even Functions

A function $f(x)$ is **even** if $f(x) = f(-x)$. Some examples are $f(x) = x^2$, $f(x) = |x|$, and, of particular importance in our case, $f(x) = \cos(x)$.

The graph of an even function looks the same when reflected in the vertical axis $x = 0$.

A function $f(x)$ is **odd** if $f(x) = -f(-x)$. Some examples are $f(x) = x^3$, $f(x) = x$, and, of particular importance in our case, $f(x) = \sin(x)$.

The graph of an odd function looks the same when rotated about the origin by π radians.

Identities

Using the properties of odd and even functions, the first two identities (integrals of even functions) may also be written:

$$\blacktriangleright \frac{2}{L} \int_0^L \sin\left(\frac{m\pi}{L}x\right) \sin\left(\frac{n\pi}{L}x\right) dx = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

$$\blacktriangleright \frac{2}{L} \int_0^L \cos\left(\frac{m\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) dx = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

for integers m and n . Also,

$$\cos(2\theta) = 1 - 2\sin^2(\theta) = 2\cos^2(\theta) - 1$$

so

$$\cos^2\left(\frac{n\pi x}{L}\right) = \frac{\cos\left(\frac{2n\pi x}{L}\right) + 1}{2} \quad \text{and} \quad \sin^2\left(\frac{n\pi x}{L}\right) = \frac{1 - \cos\left(\frac{2n\pi x}{L}\right)}{2}$$

Fourier Series

When analysing a periodic function with a Fourier series, we only need to focus on one period.

Example: Find the Fourier series of the function:

$$f(x) = \begin{cases} -k & \text{if } -\pi < x \leq 0 \\ k & \text{if } 0 < x \leq \pi \end{cases}$$

and $f(x + 2\pi) = f(x)$, a square wave.

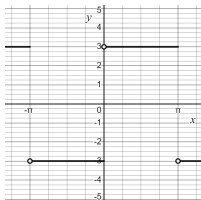


Figure: Setting $k = 3$

Fourier Series Calculations

Find a_0 .

$$\begin{aligned}a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx \\&= \frac{1}{2\pi} \int_{-\pi}^0 -k \, dx + \frac{1}{2\pi} \int_0^{\pi} k \, dx \\&= \frac{-kx}{2\pi} \Big|_{-\pi}^0 + \frac{kx}{2\pi} \Big|_0^{\pi} \\&= 0 + \frac{-k\pi}{2\pi} + \frac{k\pi}{2\pi} + 0 \\&= 0\end{aligned}$$

Since the average height of the function is zero we would expect $a_0 = 0$.

Fourier Series Calculations

Find a_n , $n \geq 1$.

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx \\&= \frac{1}{\pi} \int_{-\pi}^0 -k \cos(nx) \, dx + \frac{1}{\pi} \int_0^{\pi} k \cos(nx) \, dx \\&= \frac{-k}{n\pi} \sin(nx) \Big|_{-\pi}^0 + \frac{k}{n\pi} \sin(nx) \Big|_0^{\pi} \\&= 0\end{aligned}$$

Since the function is odd we would expect $a_n = 0$.

Fourier Series Calculations

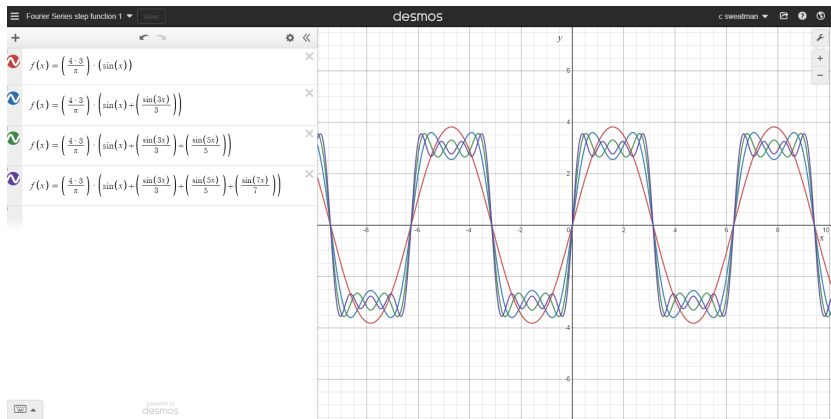
Find b_n , $n \geq 1$.

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx \\ &= \frac{1}{\pi} \int_{-\pi}^0 -k \sin(nx) \, dx + \frac{1}{\pi} \int_0^{\pi} k \sin(nx) \, dx \\ &= \frac{k}{n\pi} \cos(nx) \Big|_{-\pi}^0 + \frac{-k}{n\pi} \cos(nx) \Big|_0^{\pi} \\ &= \frac{k}{n\pi} (1 - \cos(-n\pi)) - \frac{k}{n\pi} (\cos(n\pi) - 1) \\ &= \frac{2k}{n\pi} (1 - \cos(n\pi)) \end{aligned}$$

So $b_n = 0$ if n is even and $b_n = \frac{4k}{n\pi}$ if n is odd. So

$$f(x) = \frac{4k}{\pi} \sum_{n=0}^{\infty} \frac{\sin((2n+1)x)}{(2n+1)} = \frac{4k}{\pi} \left(\sin(x) + \frac{\sin(3x)}{3} + \frac{\sin(5x)}{5} + \frac{\sin(7x)}{7} + \dots \right)$$

Approximating Square Wave with $k = 3$



Fourier Series

Of course, in general, periodic functions can have any period. For a function $f(x)$ of period $2L$, we can represent it as follows:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \right)$$

where:

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) \, dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L}x\right) \, dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) \, dx$$

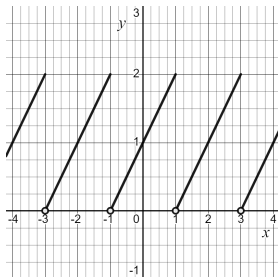
$n \geq 1$.

Fourier Series

Example: Find the Fourier series of the function defined by:

$$f(x) = x + 1 \quad \text{for } -1 < x \leq 1$$

and $f(x + 2) = f(x)$, a sawtooth function.



Fourier Series Calculations

Here $L = 1$. Find a_0 .

$$\begin{aligned}a_0 &= \frac{1}{2} \int_{-1}^1 f(x) \, dx \\&= \frac{1}{2} \int_{-1}^1 (x + 1) \, dx \\&= \frac{1}{4} (x + 1)^2 \Big|_{-1}^1 \\&= 1\end{aligned}$$

Since the average height of the function is one we would expect $a_0 = 1$.

Note that f is an odd function to which one has been added.

Fourier Series Calculations

Find a_n , $n \geq 1$.

$$\begin{aligned}a_n &= \int_{-1}^1 (x+1) \cos(n\pi x) \, dx \\&= \frac{(x+1)}{n\pi} \sin(n\pi x) \Big|_{-1}^1 - \frac{1}{n\pi} \int_{-1}^1 \sin(n\pi x) \, dx \\&= 0 - 0 + \frac{\cos(n\pi x)}{(n\pi)^2} \Big|_{-1}^1 \\&= \frac{\cos(n\pi)}{(n\pi)^2} - \frac{\cos(-n\pi)}{(n\pi)^2} \\&= 0\end{aligned}$$

Since the function minus its average value is odd we would expect $a_n = 0$. We integrated by parts using $u = (x+1)$ and $v = \frac{\sin(n\pi x)}{n\pi}$ and

$$\int u \, dv = uv - \int v \, du$$

Fourier Series Calculations

Find b_n , $n \geq 1$.

$$\begin{aligned} b_n &= \int_{-1}^1 (x+1) \sin(n\pi x) \, dx \\ &= \frac{(x+1)}{n\pi} (-\cos(n\pi x)) \Big|_{-1}^1 + \frac{1}{n\pi} \int_{-1}^1 \cos(n\pi x) \, dx \\ &= \frac{(-2)}{n\pi} \cos(n\pi) + 0 + \frac{\sin(n\pi x)}{(n\pi)^2} \Big|_{-1}^1 \\ &= \frac{(-2)}{n\pi} \cos(n\pi) \end{aligned}$$

So, $b_n = \frac{-2}{n\pi}$ if n is even and $b_n = \frac{2}{n\pi}$ if n is odd. We integrated by parts using $u = (x+1)$ and $v = -\frac{\cos(n\pi x)}{n\pi}$ and

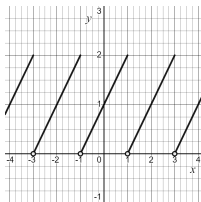
$$\int u \, dv = uv - \int v \, du$$

Fourier Series

Hence, for the function defined by:

$$f(x) = x + 1 \quad \text{for } -1 < x \leq 1$$

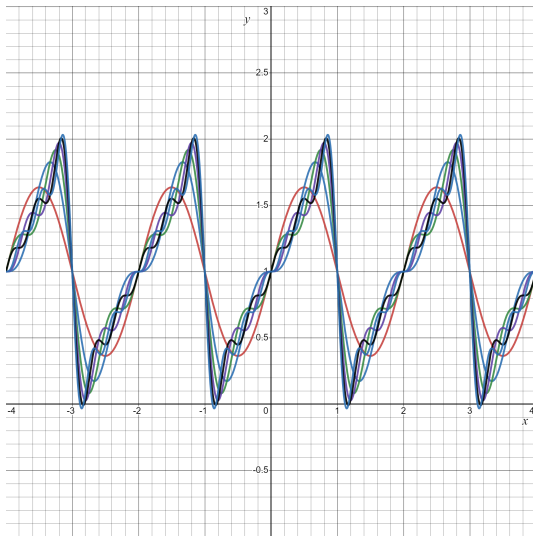
$$\text{and } f(x+2) = f(x)$$



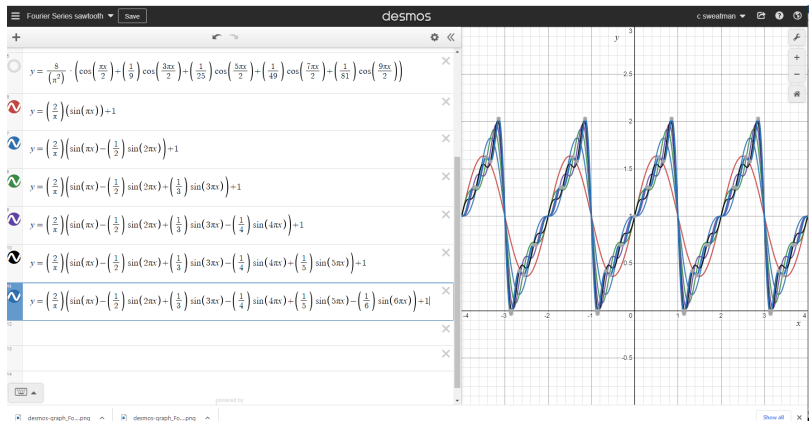
the Fourier series is

$$f(x) = 1 + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin(n\pi x)}{n}$$

Adding Sine Functions



Adding Sine Functions



Fourier Series: Odd and Even Functions

If a periodic function is even, in its Fourier series, all the b_i terms will be 0, that is, it will only contain $\cos$ terms.

If a periodic function is odd, in its Fourier series, all the a_i terms will be 0, that is, it will only contain $\sin$ terms.

In both examples above, the functions are odd (the second one after we have shifted it down by the constant 1), hence their Fourier series only contained $\sin$ terms.

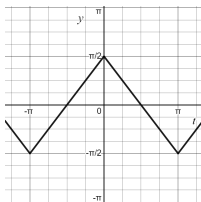
Sometimes (later in the course) we have a function defined on the interval $[0, L]$. We can extend this function to the interval $[-L, L]$, and then extend to the real line, as either even or odd, depending on whether we need a sum of sines or a sum of cosines.

Fourier Series

Exercise: Which terms will be included in the Fourier series of the function defined by:

$$f(t) = \begin{cases} \frac{\pi}{2} + t & \text{if } -\pi < t \leq 0 \\ \frac{\pi}{2} - t & \text{if } 0 < t \leq \pi \end{cases}$$

and $f(t + 2\pi) = f(t)$?



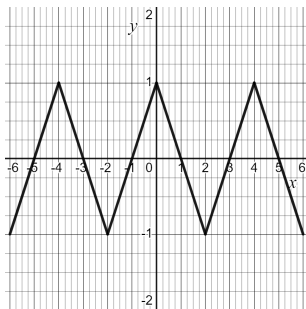
Fourier Series

Exercise:

- Find the Fourier series of the function defined by:

$$f(x) = \begin{cases} 1 + x & \text{if } -2 < x < 0 \\ 1 - x & \text{if } 0 < x < 2 \end{cases}$$

and $f(x + 4) = f(x)$.



- Check your calculations by graphing the sum of the first few terms.

Text Book References

Chapters 11.1, 11.2